DO NOW — RECALL

Lesson 052 · Binary Search Trees with Node Objects

Name: _____ Class: _____ Date: _____

1. What does LinkedList<T>.AddAfter use to place a new node?

2. What does HEAD mean for a linked list?

3. What is an invariant of a BST?

4. A binary search tree has root 10, left child 6 and right child 18. Insert 15, then 12, using the usual smaller-left, larger-right rule. Determine the parent and side of the new node 12.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

An existing node reference.

2. Model answer

Its first node.

3. Model answer

Left subtree keys less and right keys greater under chosen policy.

4. Model answer

12 is the left child of 15. Its search path is right from 10, left from 18, then left from 15.

By the end, you can…

• A binary search tree (BST) keeps smaller keys in the left subtree and larger keys in the right subtree under a consistent duplicate-key policy.

• Each node stores a key and optional Left/Right child references. Inserting begins at the root and follows comparisons until an empty child link is found.

• Inorder traversal (left, root, right) prints sorted keys for a valid BST without duplicates.

Vocabulary — know these terms

BST — Binary search tree ordered by key.

root — Topmost tree node.

leaf — Node without child nodes.

inorder — Left subtree, root, right subtree.

NEW LEARNING

Read, annotate and explain

Discover

A binary search tree (BST) keeps smaller keys in the left subtree and larger keys in the right subtree under a consistent duplicate-key policy.

Trace

Each node stores a key and optional Left/Right child references. Inserting begins at the root and follows comparisons until an empty child link is found.

Evaluate

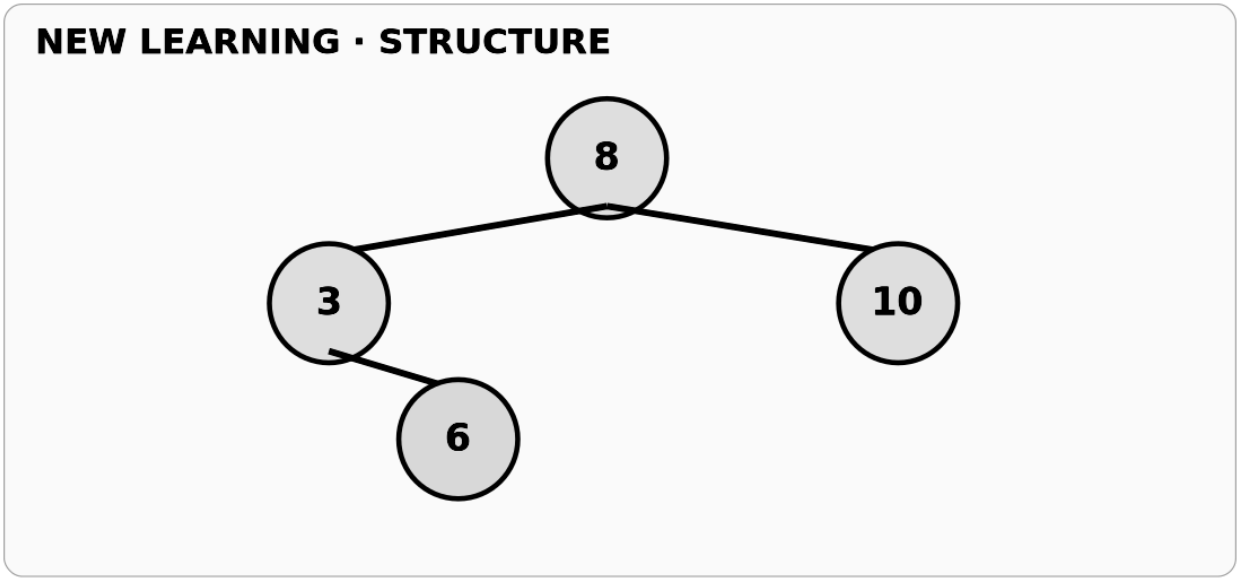
Inorder traversal (left, root, right) prints sorted keys for a valid BST without duplicates.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · STRUCTURE: 8, 3, 10, 6

I DO — WORKED EXAMPLE

Follow the worked method

Insert key 7 into BST root 8, left 3, right 10, with 6 right of 3.

Explain the trace

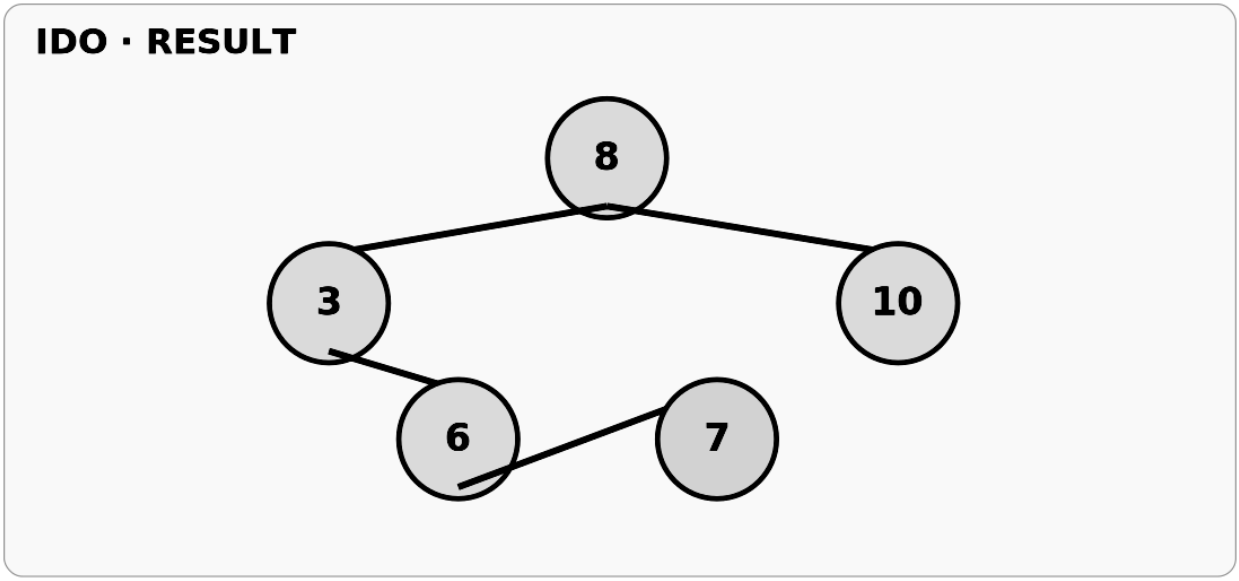
7 becomes right child of 6; follow comparisons until a null child is reached.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert key 7 into BST root 8, left 3, right 10, with 6 right of 3.

7 becomes right child of 6; follow comparisons until a null child is reached.



IDO · RESULT: 8, 3, 10, 6, 7

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Insert 12 into a different BST: root 9, children 5 and 14.
Source code / input:
12>9 → right(14)
12<14 → left child of 14

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

BST root8, left3, right10, insert 6. Show comparisons and location. [6]

Question 2

For BST with keys 8,3,10,6, give inorder traversal and explain sorting. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Explain why BST search may be O(log n) when balanced but O(n) when skewed. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

An unbalanced BST is guaranteed to have logarithmic search time.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

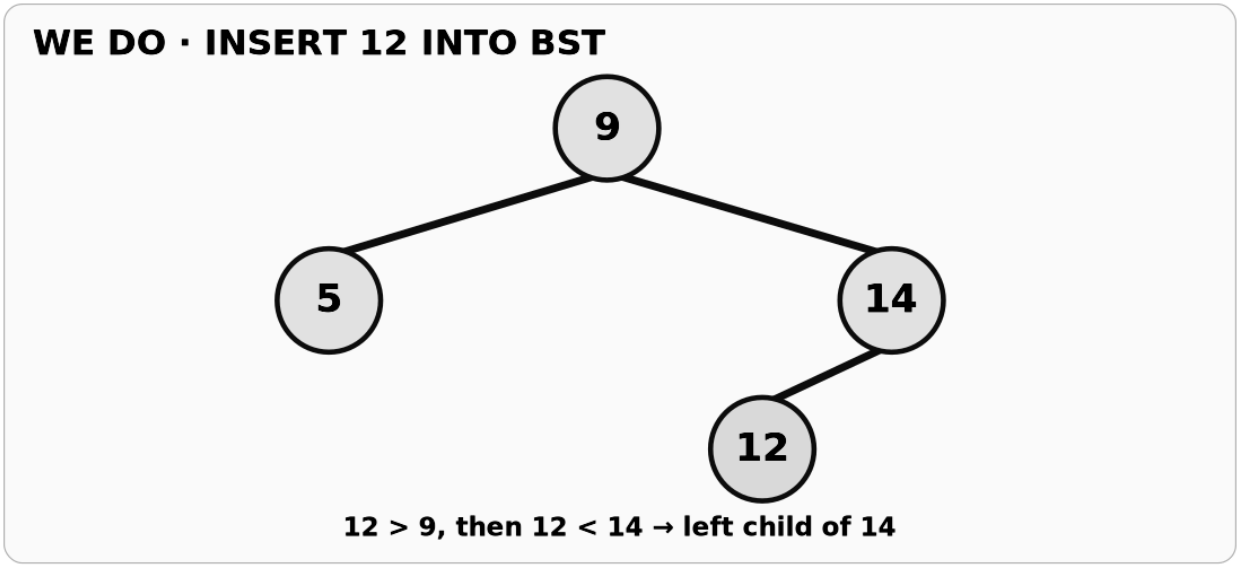
Explain the trace
Place 12 as left child of 14. Tree ordering remains valid.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 12 into a different BST: root 9, children 5 and 14.

Place 12 as left child of 14. Tree ordering remains valid.



WE DO BST: root 9, left child 5, right child 14, key 12 inserted as left child of 14